

Chapter 1: Introduction

Spatial Econometrics

Spatial econometrics

Spatial econometrics concerns spatial dependence or social interaction among the geographical units, economic agents or social actors, such as neighbourhood effects, spillover effects, copy-catting, and peer effects.

- It has received an increased attention by regional scientists, economists, econometricians, and statisticians.
- Standard econometric techniques often fails in the presence of spatial interactions. Spatial econometric models and methods extend the standard ones to capture the spatial interactions.
- They have been applied not only in specialized fields such as regional science, urban economics, real estate and economic geography,
- but also increasingly in more traditional fields of economics, including demand analysis, labour economics, public economics, international economics, and agricultural and environmental economics.
- **And to finance as well**, for issues, e.g., asset pricing with spatial interaction (Kou, et al., 2017), stock market comovements, and more.

Course Contents

This course introduces common spatial econometric models, obtained by extending the classical linear regression models, linear panel data models, and linear dynamic panel data models, namely,

- 1 spatial linear regression (SLR) models,
- 2 spatial panel data (SPD) models,
- 3 spatial dynamic panel data (SDPD) models,

where one or more of the following features:

- spatial error dependence,
- spatial lag dependence, and
- spatial Durbin effect,

is built into the traditional models to capture the 'cross-sectional' or **spatial dependence**, resulted from the 'interactions' among the geographical units, economic agents or social actors.

Spatial dependence is modeled by a **spatial weight matrix** or a **connectivity matrix**, and a **spatial parameter**.

For each type model,

- common inference methods, such as quasi-maximum likelihood (QML), M-estimation, and GMM, are introduced,
- empirical illustrations are presented, and
- Matlab routines are provided with which students can conduct their own empirical analyses.

If time permits, recent advances on refined and robust inferences methods will be discussed.

Chapter 1 introduces common spatial econometric models and common spatial weight matrices, presents some background knowledge on QML, M-, and GMM estimation methods, and discuss some popular applications of spatial models and recent developments in spatial software tools.

Brief History of Spatial Econometrics

The term *spatial econometrics* was coined by **Jean Paelinck**, a Belgium Econometrician, in the early 1970s to designate a growing body of the regional science literature that dealt primarily with estimation and testing problems encountered in the implementation of multiregional econometric models. See the Book: Anselin (1988, p.7).

The dependence among the cross-sectional units can be considered to lie at the core of the disciplines of the regional science and geography, as expressed in Tobler's (1979) *first law of geography*:

"everything is related to everything else, but near things are more related than distant things."

See also the books of Paelinck and Klaassen (1979), LeSage and Pace (1999), and Elhorst (2014).

While the concept of spatial dependence originated from the notion of relative space or relative location, which emphasizes the effect of distance or neighborhood, the notation of space or location can easily be extended beyond the strict Euclidean space inducing

- economic distance,
- policy distance,
- inter-personal distance,
- financial networks,
- social networks, etc.

Hence, spatial dependence is a phenomenon that exists in a wide range of applications in the social sciences and economics.

Spatial Linear Regression (SLR) Models

The SLR models extend the classical linear regression models:

$$Y_n = X_n\beta + u_n, \quad (1)$$

by adding one or more of the following terms:

- 1 **spatial lag** (SL) term: $\lambda_1 W_{1n} Y_n$,
- 2 **spatial error** (SE) term: $u_n = \lambda_2 W_{2n} u_n + v_n$,
- 3 **spatial Durbin** (SD) term: $W_{3n} X_n^* \lambda_3$,

where

- Y_n ($n \times 1$), X_n ($n \times p$), β , and u_n have the usual interpretations,
- X_n^* : a submatrix of X_n excluding the column of ones, etc,
- W_{rn} , $r = 1, 2, 3$: known $n \times n$ spatial weight matrices, and
- λ_1 , λ_2 , and λ_3 are the parameters corresponding to, respectively, the SL, SE, and SD effects.

Spatial Panel Data (SPD) Models

The SPD models extend the classical panel data models:

$$Y_{nt} = X_{nt}\beta + Z_n\gamma + \mu_n + \alpha_t\mathbf{1}_n + u_{nt}, \quad t = 1, \dots, T, \quad (2)$$

by adding one or more of the following: $\lambda_1 W_{1n} Y_{nt}$, $\lambda_2 W_{2n} u_{nt}$, and $\lambda_3 W_{3n} X_{nt}^*$, being, respectively, the SL, SE and SD effects, where

- X_{nt} : $n \times p$ matrix or time-varying regressors,
- Z_n : $n \times q$ matrix of time-invariant regressors,
- μ_n is an $n \times 1$ vector of unit-specific effects,
- $\{\alpha_t\}$ are the time-specific effects,
- and the other quantities are similarly defined.

The μ_n and $\{\alpha'_t s\}$ are called: (i) **fixed effects** if they are allowed to be correlated with the time-varying regressors X_{nt} in an arbitrary manner; (ii) **random effects** if they are uncorrelated with X_{nt} ; or (iii) **correlated random effects** if they are correlated with X_{nt} linearly.

Spatial Dynamic Panel Data (SDPD) Models

The SDPD models extend the classical dynamic panel data models:

$$Y_{nt} = \rho Y_{n,t-1} + X_{nt}\beta + Z_n\gamma + \mu_n + \alpha_t\mathbf{1}_n + u_{nt}, \quad t = 1, \dots, T, \quad (3)$$

by adding the SL, SE, or SD terms as in the SPD models. It can be further extended by adding a space-time effect $\lambda W_n Y_{n,t-1}$.

Challenges with the SDPD model:

- 1 it incurs the **incidental parameters problem** (number of parameters increase with the increase of sample size),
- 2 it gives rise to the **initial values problem** (the distribution of the initial observations Y_{n0} depends on the past values which are unobservables).

Remarks: For all the spatial econometric models, the spatial weight matrices can be the same. The SL and SE effects can go for higher order.

Spatial Weight Matrix

Spatial weight matrix $W_n = \{w_{ij}, i, j = 1, \dots, n\}$ is an $n \times n$ matrix that measures the 'connectivity' among the n spatial units. The construction of W_n is a subject matter, depending on the actual problem considered.

Contiguity-based Weights: $w_{ij} = 1$ when units i and j are 'neighbors', and $w_{i,j} = 0$ otherwise. Popular ones include:

- **Rook, Bishop, or Queen Contiguity:** in a regular grid, the neighbors are those with common edge, or common vertex, or combination of both, in analogy with the move of Rook, Bishop or Queen in the game of chess (Anselin, 1988, p.18);
- **Circular World:** units immediate ahead or behind of a given spatial unit are considered as neighbors of this unit, e.g., "5 ahead and 5 behind" (Kelejian and Prucha, 1999, p. 520);
- **Group Interaction:** when n spatial units are naturally from R groups, the members within each group are neighbors, but the members from different groups are not (Case, 1991; Lee, 2004).

Weights based on Physical Distance: The concept of neighbors can be extended to mean areal units with (i) common border, (ii) within a given centroid distance of each other, or (iii) closest in terms of centroid distance, etc., resulting W_n with elements 1 or 0 as above:

- **Common Border Weights:** $w_{ij} = 1$ if units i and j share common border, and 0 otherwise;
- **K-Nearest Neighbors Weights:** $w_{ij} = 1$ if units $j (= 1, \dots, K)$ are the K units closest to the unit i in terms of centroid distance;
- **Radial Distance Weights:** $w_{ij} = 1$ for $d_{ij} \leq \delta$, where d_{ij} is the distance between units i and j , and δ is the distance cutoff value, giving the distance-based contiguity.

Spatial weights w_{ij} can also be defined directly in terms of the actual distance d_{ij} between units i and j , or in terms of the length ℓ_{ij} of the common boarder between units i and j :

- Power Distance Weights: $w_{ij} = d_{ij}^{-\alpha}$, where α is any positive exponent;
- Exponential Distance Weights: $w_{ij} = \exp(-\alpha d_{ij})$, where α is a positive exponent;
- Shared-Boundary Weights: $w_{ij} = \ell_{ij} / \sum_{k \neq i} \ell_{ik}$, proportion of $i - j$ boundary in the total boundary of unit i .

Weights based on Social or Economic Distance: Other specification of spatial weights are possible as well (Anselin and Bera, 1998., p.244).

- In sociometrics, the weights reflect whether or not two individual belong to the same social network (Doreian, 1980).
- In **economics applications**, the use of spatial weights based on “economic” distance has been suggested, among others, in Case et al. (1993). Specifically, they suggest to use weights of the form

$$w_{ij} = 1/|x_i - x_j|,$$

where x_i and x_j are observations on “meaningful” socioeconomic characteristics such as per capita income or percentage of the population in a given racial or ethnic group.

The final form of W_n is such that $w_{ii} = 0$, and W_n is row normalized, i.e., each element is divided by its row sum.

Possible endogeneity in spatial weight matrix.

Anselin and Bera (1998, p.244) made an important note:

in the standard estimation and testing problems, the weights matrix is taken to be exogenous. Therefore, indicators for the socioeconomic weights should be chosen with great care to ensure their exogeneity, unless their endogeneity is considered explicitly.

Roles of Spatial Terms

Consider a spatial contiguity-based weight matrix with $n = 12$ spatial units:

$$W_n = \begin{pmatrix} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/4 & 0 & 1/4 & 1/4 & 1/4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 0 & 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/4 & 1/4 & 0 & 0 & 0 & 0 & 0 & 0 & 1/4 & 1/4 & 0 \\ 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 0 & 1/3 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \end{pmatrix}$$

Spatial Error Mode: $Y_n = X_n\beta + u_n$, $u_n = \lambda W_n u_n + v_n$.

$$\Rightarrow u_1 = (u_6 + u_7)/2 + v_1; \quad u_4 = (u_3 + u_5 + u_{10})/3 + v_4, \text{ etc.}$$

In general, $u_n = (I_n - \lambda W_n)^{-1} v_n$.

- If $\{v_i\} \stackrel{iid}{\sim} (0, \sigma_v^2)$, then $\text{Var}(u_n) = \sigma_v^2 [(I_n - \lambda W_n)'(I_n - \lambda W_n)]^{-1}$.
 $\Rightarrow \text{Var}(Y_n) = \sigma_v^2 [(I_n - \lambda W_n)'(I_n - \lambda W_n)]^{-1}$.
- However, $E(Y_n) = X_n \beta$, the same as the regular linear model.

Spatial error induces **cross-sectional dependence** in the disturbances u_n , and thus changes the variance-covariance of the response Y_n , but does not induce changes in the mean level of Y_n .

Spatial Lag Model: $Y_n = \lambda W_n Y_n + X_n \beta + u_n$.

- For example, $Y_1 = (Y_6 + Y_7)/2 + x_1' \beta + u_1$, etc.
- $Y_n = (I_n - \lambda W_n)^{-1} (X_n \beta + u_n)$.
- If $\{u_i\} \stackrel{iid}{\sim} (0, \sigma_u^2)$, then $E(Y_n) = (I_n - \lambda W_n)^{-1} X_n \beta$,
and $\text{Var}(Y_n) = \sigma_u^2 [(I_n - \lambda W_n)'(I_n - \lambda W_n)]^{-1}$.

Spatial lag induces the so-called cross-sectional dependence in u_n and thus in Y_n , and also changes the mean of Y_n .

Spatial Durbin Model: e.g., $Y_n = \beta_0 \mathbf{1}_n + X_n \beta_1 + \lambda W_n X_n \gamma + u_n$.

- For example, $Y_1 = \beta_0 + x_1 \beta_1 + [(x_6 + x_7)/2] \gamma + u_1$.

The mean of the response Y_1 of the spatial unit 1 is affected not only by the value of its own regressor x_1 , but also by the regressor values of the neighboring spatial units 6 and 7, and similarly for the means of the responses of the other spatial units.

- The term **Spatial Durbin** first appeared in Anselin (1988, p. 40, p. 111), for its analogy to the suggestion by Durbin (1960) for the time series case.
- SD term induces the **local effects** (from the immediate neighbors), in contrast to the SL effect which induces the **global effects** (from the immediate neighbors, neighbors of the immediate neighbors, etc.).
- Change in, e.g., Y_1 caused by the change in x_1 is referred to as the **direct effect**; and changes in other Y 's caused by the change in x_1 are referred to as the **indirect effects** that could be local, or global.

Maximum Likelihood Estimation

The maximum likelihood (ML) estimator holds special place among estimators (Cameron and Trivedi, 2005, p.139), as

- it is the most efficient estimator among estimators which are **consistent** and **asymptotically normal**, and
- the principle behind the ML estimation lays out a foundation based on which many other more general methods are obtained such as **quasi-ML estimation** and **M-estimation**.

The **likelihood principle**, due to a pioneer statistician R. A. Fisher (1922), is to choose, as an estimator of parameter vector θ_0 , a value of θ that maximizes the **likelihood** of observing the actual sample data.

The joint probability density function (pdf) or joint probability mass function (pmf) of random observations is the 'likelihood' for a particular sample to be observed.

Let $f(\mathbf{y}|\theta)$ be the joint pdf of random vector Y indexed by a vector θ of unknown parameters, given the exogenous regressors values X and exogenous spatial weights matrix W .

- Now, $f(\mathbf{y}|\theta)$ is viewed as a function of θ and is denoted by $L(\theta|\mathbf{y})$ or simply $L_n(\theta)$.
- Then $L_n(\theta)$ is called the **likelihood function** of θ based on the observed vector $\mathbf{y}$ on Y .
- Maximizing $L_n(\theta)$ is equivalent to maximizing the loglikelihood $\ell_n(\theta) = \ln L_n(\theta)$, and

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} \ell_n(\theta), \quad (4)$$

is the so-called **maximum likelihood estimator** (MLE) of θ , where Θ denotes the parameter space.

- A simple interpretation could be: the observed vector $\mathbf{y}$ most likely 'came' from the 'population' represented by $f(\mathbf{y}|\hat{\theta}_n)$.

There are some important properties of the ML principle.

- Let $S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta)$, then, $S_n(\theta)$ is called **score function**.
- At the **true value** θ_0 of θ , $S_n(\theta_0)$ is called **efficient score**.
- For standard ML estimation problems, $\hat{\theta}_n$ defined in (4) is equivalent to

$$\hat{\theta}_n = \arg\{S_n(\theta) = 0\}. \quad (5)$$

- Let 'E' denote the expectation with respect to the true pdf $f(\mathbf{y}|\theta_0)$, then

$$E[S_n(\theta_0)] = 0, \quad (6)$$

- The expectation of the outer product of the efficient score vector:

$$\mathcal{I}_n \equiv \mathcal{I}_n(\theta_0) = E[S_n(\theta_0)S_n'(\theta_0)], \quad (7)$$

is called the **Fisher Information**, commonly known as the **information matrix** or the **expected information matrix**.

The well-known **information matrix equality (IME)** holds under true $\ell(\theta)$:

$$-E[H_n(\theta_0)] = E[S_n(\theta_0)S_n'(\theta_0)], \quad (8)$$

where $H_n(\theta_0) = \frac{\partial}{\partial \theta'} S_n(\theta_0)$, called the **Hessian** matrix. The negative Hessian matrix is referred to as the **Observed Information Matrix**.

The proof of (6) goes as follows:

$$\begin{aligned} E[S_n(\theta_0)] &= \int S_n(\theta_0) f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= \int \frac{\partial \ell_n(\theta_0)}{\partial \theta} f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= \int \frac{\partial \ln f(\mathbf{y}|\theta_0)}{\partial \theta} f(\mathbf{y}|\theta_0) d\mathbf{y} = \int \frac{\partial}{\partial \theta} f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= \frac{\partial}{\partial \theta} \int f(\mathbf{y}|\theta_0) d\mathbf{y} = \frac{\partial}{\partial \theta} (1) = 0, \end{aligned}$$

where differentiation and integration are assumed to be interchangeable, which is typically valid if the range of $\mathbf{y}$ does not depend on θ .

The proof of (8) goes as follows. Note that $H_n(\theta) = \frac{\partial^2}{\partial\theta\partial\theta'}\ell_n(\theta)$. We have

$$\begin{aligned} \mathbb{E}[H_n(\theta_0)] &= \int \frac{\partial^2 \ell_n(\theta_0)}{\partial\theta\partial\theta'} f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= \int \frac{\partial S_n(\theta_0)}{\partial\theta'} f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= \int \frac{\partial}{\partial\theta'} [S_n(\theta_0) f(\mathbf{y}|\theta_0)] d\mathbf{y} - \int S_n(\theta_0) \left[\frac{\partial}{\partial\theta'} f(\mathbf{y}|\theta_0) \right] d\mathbf{y} \\ &= \int \frac{\partial}{\partial\theta'} [S_n(\theta_0) f(\mathbf{y}|\theta_0)] d\mathbf{y} - \int S_n(\theta_0) S'_n(\theta_0) f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= \frac{\partial}{\partial\theta'} \int S_n(\theta_0) f(\mathbf{y}|\theta_0) d\mathbf{y} - \int S_n(\theta_0) S'_n(\theta_0) f(\mathbf{y}|\theta_0) d\mathbf{y} \\ &= 0 - \mathbb{E}[S_n(\theta_0) S'_n(\theta_0)] \\ &= -\mathcal{I}_n. \end{aligned}$$

Under some regularity conditions, we have

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, \lim_{n \rightarrow \infty} n\mathcal{I}_n^{-1}). \quad (9)$$

Quasi Maximum Likelihood Estimation

In practice, the true distribution of Y , say $g(\mathbf{y}|\theta)$, is often unknown, the 'chosen' distribution $f(\mathbf{y}|\theta)$ can best be an approximation to $g(\mathbf{y}|\theta)$. Thus,

- $L_n(\theta)$ defined in terms of $f(\mathbf{y}|\theta)$ is a wrong (quasi) likelihood function. Would maximizing the quasi-likelihood $L_n(\theta)$ still lead to a consistent estimator for θ_0 ? Would the asymptotic normality result still hold?
- It is well known that for a maximum likelihood type estimator, or M-estimator, to be consistent, it is necessary that

$$\text{plim} \frac{1}{n} S_n(\theta_0) = 0, \quad (10)$$

which boils down by requiring that $\lim \frac{1}{n} E[S_n(\theta_0)] = 0$.

- However, this is not guaranteed if $f(\mathbf{y}|\theta)$ is a **misspecification**.
- Very interestingly, we can show that for a **certain choice** of $f(\mathbf{y}|\theta)$ that partially specifies $g(\mathbf{y}|\theta)$ in the sense that its first two moments matches these of $g(\mathbf{y}|\theta)$, consistency can also be achieved.
- One such a choice is the Gaussian likelihood.

Let $Y \sim (\mu_n, \Sigma_n)$, i.e., the true joint pdf $g(\mathbf{y}|\theta_0)$ is known only up to the first two moments: $E(Y) = \mu_n \equiv \mu_n(\beta_0)$ and $\text{Var}(Y) = \Sigma_n \equiv \Sigma_n(\gamma_0)$.

If the specified joint pdf is a normal, i.e., $f(\mathbf{y}|\theta_0) \sim N[\mu(\beta_0), \Sigma(\gamma_0)]$, the quasi loglikelihood is

$$\ell_n(\theta) = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_n(\gamma)| - \frac{1}{2} [\mathbf{y} - \mu_n(\beta)]' \Sigma_n^{-1}(\gamma) [\mathbf{y} - \mu_n(\beta)], \quad (11)$$

where , and $\theta = (\beta', \gamma')'$. The quasi score function is

$$S_n(\theta) = \begin{cases} \dot{\mu}'_n(\beta) \Sigma_n^{-1}(\gamma) [\mathbf{y} - \mu_n(\beta)], \\ -\Sigma_n^{-1}(\gamma) \dot{\Sigma}_{nk}(\gamma) + [\mathbf{y} - \mu_n(\beta)]' \Sigma_n^{-1}(\gamma) \dot{\Sigma}_{nk}(\gamma) \Sigma_n^{-1}(\gamma) [\mathbf{y} - \mu_n(\beta)], \end{cases}$$

for $k = 1, \dots, \dim(\gamma)$, where $\dot{\mu}'_n(\beta) = \frac{\partial}{\partial \beta} \mu_n(\beta)$ and $\dot{\Sigma}_{nk}(\gamma) = \frac{\partial}{\partial \gamma_k} \Sigma_n(\gamma)$.

Clearly,

$$E[S_n(\theta_0)] = 0,$$

whether or not $f(\mathbf{y}|\theta) = g(\mathbf{y}|\theta)$, as the quasi score $S_n(\theta_0)$ is a linear-quadratic (LQ) function of $u_n = \mathbf{y} - \mu_n(\beta_0)$!

- As long as the two distribution have the same first two moments, the expectation of $S_n(\theta_0)$ equals zero.
- See Huber (1967) for the original ideas of QML estimation.
- See also White (1982) for a similar interpretation and for the related **information matrix test**.

Maximizing $\ell_n(\theta)$ gives the quasi (Gaussian) MLE, or QMLE $\hat{\theta}_n$ of θ if $f(\mathbf{y}|\theta) \neq g(\mathbf{y}|\theta)$, which become MLE if $f(\mathbf{y}|\theta) = g(\mathbf{y}|\theta)$. The Gaussian QMLE $\hat{\theta}_n$ would be still consistent and asymptotically normal. The difference is that its asymptotic variance becomes

$$\text{Var}(\hat{\theta}_n) \stackrel{a}{=} \mathcal{J}_n^{-1}(\theta_0) \mathcal{I}_n(\theta_0) \mathcal{J}_n^{-1}(\theta_0), \quad (12)$$

where $\mathcal{J}_n(\theta_0) = -E[H_n(\theta_0)]$.

- Thus, in case that the true underlying distribution of a model is nonnormal, but normal (Gaussian) likelihood is used, the resulted estimator would still be consistent and asymptotically normal. This makes the Gaussian QML estimation method very attractive.
- A **limitation** of the QML method is that it cannot be applied to models with additional endogenous regressors where endogeneity is of **implicit form**, unlike the endogenous 'spatial' regressor *WY*.
- Similar and more general results hold for any other distribution from the **exponential family** (see Cameron and Trivedi, 2005, Sec. 5.7).

M-Estimation

The term 'M-estimator' was coined by Huber (1964) to mean the **maximum likelihood type** estimator (see also Huber, 1969). An M-estimator $\hat{\theta}_n$ of a $p \times 1$ vector of parameters θ is defined in two ways:

- (a) as the solution of a maximization problem and
- (b) as the root of a set of estimating equations.

In case (a), $\hat{\theta}_n$ is defined as

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} Q_n(\theta), \quad (13)$$

For real valued, random criterion function $Q_n(\theta)$.

In case (b), $\hat{\theta}_n$ is defined as

$$\hat{\theta}_n = \arg \{ \Psi_n(\theta) = 0 \}, \quad (14)$$

for $p \times 1$ vector-valued **estimating function** $\Psi_n(\theta)$.

The asymptotic behavior of the M-estimator $\hat{\theta}_n$ is similar to that of the QML estimator. In particular, it is consistent and asymptotically normal.

Under regularity conditions, the M-estimator $\hat{\theta}_n$ is such that $\hat{\theta}_n \xrightarrow{p} \theta_0$, and

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N[0, \lim_{n \rightarrow \infty} n \Sigma_n^{-1}(\theta_0) \Gamma_n(\theta_0) \Sigma_n^{-1}(\theta_0)],$$

where $\Gamma_n(\theta_0) = \text{Var}[\Psi_n(\theta_0)]$ and $\Sigma_n(\theta_0) = -E[\frac{\partial}{\partial \theta'} \Psi_n(\theta_0)]$.

- The original formulation of the M-estimation is in terms of Q_n or Ψ_n which is the sum of n independent quantities. However, this can be generalized to allow either function to have more general forms to suit for the estimation of more general models such as the spatial econometric models considered in this course.
- See van der Vaart (1998) for more general formulations of M-estimation strategy, including the methods for proving its asymptotic properties.
- See also White (1994) for the relevant asymptotic methods.

Generalized Method of Moments Estimation

The generalized method of moments (GMM), developed by Lars Peter Hansen in 1982 as a generalization of the method of moments which was introduced by Karl Pearson in 1894, is a generic method for estimating parameters in statistical models.

- Usually it is applied in the context of semiparametric models, where the parameter of interest is finite-dimensional, whereas the full shape of the distribution function of the data may not be known, and therefore maximum likelihood estimation is not applicable.
- The method requires that certain **moment conditions** be specified for the model. These moment conditions are functions of the model parameters and the data, such that **their expectation is zero at the true values of the parameters.**
- The GMM method then minimizes a certain norm of the sample moment conditions.

In its most compact form, the GMM estimator $\hat{\theta}_n$ minimizes the following object function:

$$Q_n(\theta) = \mathbf{g}'_n(\theta)\Omega_n\mathbf{g}_n(\theta), \quad (15)$$

where $\mathbf{g}_n(\theta)$ is a $r \times 1$ vector of sample moments, θ is a $p \times 1$ vector of parameters ($p \leq r$), and Ω_n is the weights matrix.

Under certain regularity conditions, the GMM estimator $\hat{\theta}_n$ defined above is consistent, i.e., $\hat{\theta}_n \xrightarrow{P} \theta_0$, and asymptotically normal, i.e.,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N[0, \lim_{n \rightarrow \infty} n(\Sigma'_n \Omega_n \Sigma_n)^{-1} (\Sigma'_n \Omega_n \Gamma_n \Omega_n \Sigma_n) (\Sigma'_n \Omega_n \Sigma_n)^{-1}],$$

where $\Gamma_n = \text{Var}[\mathbf{g}_n(\theta_0)]$ and $\Sigma_n = -E[\frac{\partial}{\partial \theta'} \mathbf{g}_n(\theta_0)]$.

GMM estimator is efficient in the class of all estimators that do not use any extra information aside from that contained in the moment conditions.

- Choice of weights matrix Ω_n is an issue of concern.
- Choice of moment conditions is another issue of concern.

Optimal GMM. If Γ_n is 'known', then one can choose $\Omega_n = \Gamma_n^{-1}$. In this case, the asymptotic variance-covariance (VC) matrix of the GMM estimator simplifies to

$$(\Sigma_n \Gamma_n^{-1} \Sigma_n)^{-1},$$

a similar form to the asymptotic VC matrix of an M-estimator.

When $r = p$, the model is said to be **just-identified**. In this case, Ω_n is simply taken to be the identity matrix, and the GMM estimator becomes the **method of moments** (MM) estimator defined as

$$\hat{\theta}_n = \arg\{\mathbf{g}'_n(\theta) = 0\}. \quad (16)$$

Clearly, the MM estimator is also the M-estimator defined earlier. In this case, Σ_n is also invertible, and the asymptotic VC matrix can be written as $\Sigma_n^{-1} \Gamma_n \Sigma_n^{-1}$.

Neighborhood Crime. In illustrating the applications of spatial cross-sectional models, Anselin (1988, p.187) used the neighborhood crime data corresponding to 49 contiguous neighborhood in Columbus, Ohio, in 1980. These neighborhood correspond to census tracts, or aggregates of a small number of census tracts, where

- **Crime:** the combined total of residential burglaries and vehicle thefts per thousand household in the neighborhood (the response variable).
- **Income** and **House:** the explanatory variables representing income and housing values in thousand dollars.
- **East:** a dummy variable indicates whether the 'neighborhood' in the east or west of a main north-south transportation axis.
- In addition, the neighborhood centroid coordinates (X and Y) are also given, as well as the list of neighbors of each spatial unit (neighborhood) that gives a first-order contiguity spatial weight matrix.

Boston House Price. The data, given by Harrison and Rubinfeld (1978), and corrected and augmented with longitude and latitude by Gilley and Pace (1996), contains 506 observations (1 observation per census tract) from Boston Metropolitan Statistical Area, and can be found in R (`sedep`). The response variable **MEDV** and 13 explanatory variables are:

- **MEDV:** median value (corrected) of owner-occupied homes in 1000's;
- **crime:** per capita crime rate by town;
- **zoning:** proportion of residential land zoned for lots over 25,000 square feet;
- **industry:** proportion of non-retail business acres per town;
- **charlesr:** Charles River dummy variable (= 1 if tract bounds river);
- **nox:** nitric oxides concentration (parts per 10 million);
- **room:** average number of rooms per dwelling;
- **houseage:** proportion of owner-occupied units built prior to 1940;
- **distance:** weighted distances to five Boston employment centres;
- **access:** index of accessibility to radial highways;

- **taxrate:** full-value property-tax rate per 10,000;
- **ptratio:** pupil-teacher ratio by town;
- **blackpop:** $1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town;
- **lowclass:** lower status of the population proportion.

The spatial weight matrix is constructed using the Euclidean distance in terms of longitude and latitude. A threshold distance, e.g., 0.05, is chosen, which gives a W_n matrix with 19.08% non-zero elements.

Public Capital Productivity. The data, from Munnell (1990), gives indicators related to public capital productivity for 48 US states observed over 17 years (1970-1986). It can be downloaded from:

<http://pages.stern.nyu.edu/~wgreene/Text/Edition6/tablelist6.htm>

It has been extensively used for illustrating the applications of the regular panel data models, as well as recently the spatial panel data models.

In Munnell (1990), the empirical model specified is a Cobb-Douglas production function of the form:

$$\ln Y = \beta_0 + \beta_1 \ln K_1 + \beta_2 \ln K_2 + \beta_3 \ln L + \beta_4 \text{Unemp} + \epsilon,$$

with state specific fixed effects, where

- Y is the gross social product of a given state,
- K_1 is public capital,
- K_2 is private capital,
- L is labour input and Unemp is the state unemployment rate.
- The spatial weights matrix W takes a contiguity form with its (i, j) th element being 1 if states i and j share a common border, otherwise 0. The final W is row normalised.
- For models with more than one spatial term, the corresponding W 's are taken to be the same.

This model can be extended by adding the dynamic effect and one or more spatial effects.

Cigarette Demand. This is another well known panel data that has been applied under various panel data model frameworks, non-spatial or spatial, fixed effects or random effects, static or dynamic. In particular, the demand equations for cigarettes for United States were estimated, based on a panel of 46 states over 30 time periods (1963-1992), given as CIGAR.TXT on the Wiley web site associated with book of Baltagi (2013).

- Y = Cigarette sales in packs per capita (the response variable).
- X_1 = Price per pack of cigarettes;
- X_2 = Population (Pop);
- X_3 = Population above the age of 16;
- X_4 = Consumer price index with (1983=100);
- X_5 = Per capita disposable income;
- X_6 = Minimum price in adjoining states per pack of cigarettes.
- Several time dummies corresponding to the major policy interventions in 1965, 1968 and 1971 can be added into the model.
- Spatial weights can be the first-order contiguity matrix.

- The Matlab codes provided in this course are sufficient for the applications of common spatial econometrics models.
- **GeoDa** (<https://spatial.uchicago.edu/geoda>) is a free software package that conducts spatial data analysis, geovisualization, spatial autocorrelation and **spatial modeling**.
- **spdep** (<https://cran.r-project.org/web/packages/spdep/index.html>) is an R package containing a collection of R functions for spatial analyses ([spdep.pdf](#), [book.pdf](#)).
- The website: <https://spatial-panels.com/software/>, lists some spatial software codes for spatial panel data modelling.
- See the “[Handbook of Applied Spatial Analysis: Software tools, Methods and Applications](#)” for a comprehensive coverage on spatial software, methods and applications.

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